

REST

by THE
SIMPSONS



REPRESENTATIONAL STATE TRANSFER

Roy Thomas Fielding (2000)



RESOURCE ORIENTED

EEASY & SIMPLE

STANDARD & **S**TATELESS

TECHNOLOGY INDEPENDENT

Stateless = Sessionless $\neq$ Cookieless

REPRESENTATIONAL STATE TRANSFER

Roy Thomas Fielding (2000)



A "real" Web service is a service designed to embrace the "Web" from the ground up.

The Web has become the most successful distributed computing platform ever developed, thanks to years of investment and usage.

REST is not tied to any particular technology or platform – it's simply a way to design things to work like the Web.

Open Letter to W3C on Web of Services for Enterprise Computing

Nick Gall (VP Gartner)



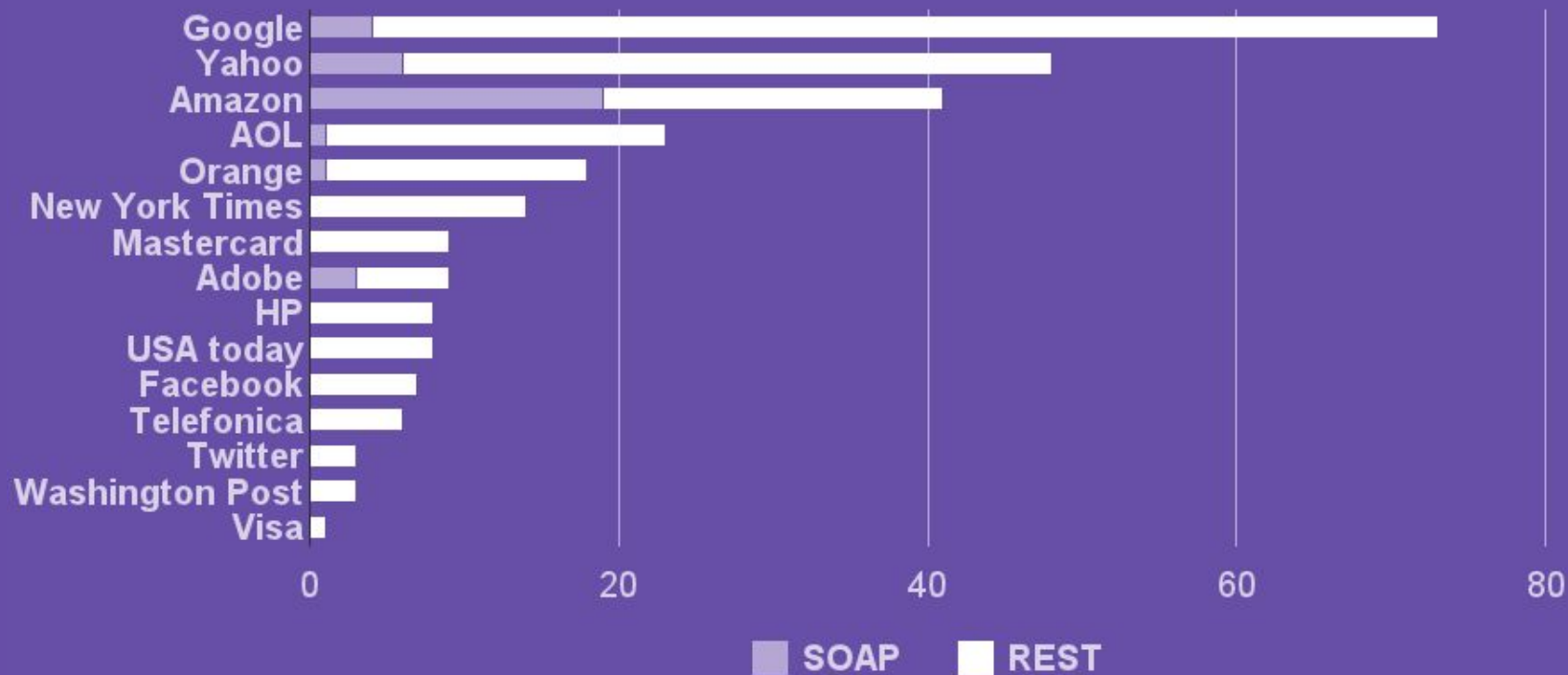
The W3C should leave the work on standardizing the WS-* middleware architecture to the middleware \$vendors\$ and shift its focus to standardizing aspects of Web architecture that make it easier to apply to "application to application" scenarios.

Traditional middleware styles include **RPC** (Remote Procedure Call, eg **DCE**), **MOM** (Message-Oriented Middleware, eg **MQSeries**), and **DOM** (Distributed Object Model, eg **CORBA** and **DCOM**). All are in production at many major enterprises. But all have fallen far short of initial expectations. Most have resulted in complex, tightly coupled, and difficult to extend integration networks.

Unfortunately, Web Services, at least the **WS-*** style, are "Web" in name only. While WS-* enables tunneling over HTTP (used merely as an XML message transport), in almost every important aspect, WS-* violates (or at best ignores) the architectural principles of the Web.

REST USAGE

AND THE REAL WORLD



¿Quién es Roy Fielding?

Un tipo inteligente.

Ayudó a hacer los primeros servidores web, se dedicó a la investigación y colaboró en la definición del protocolo sobre el que funciona la web.

edna



hawkins



¿Cómo funciona la web?

Eh... es bastante sorprendente. Y lo gracioso es que está muy subestimada.

El protocolo de la web, HTTP, es capaz de muchas cosas que la gente ignora.

http, ¿lo que escribo en el navegador?

Sí, es el protocolo más importante de la historia de la informática...

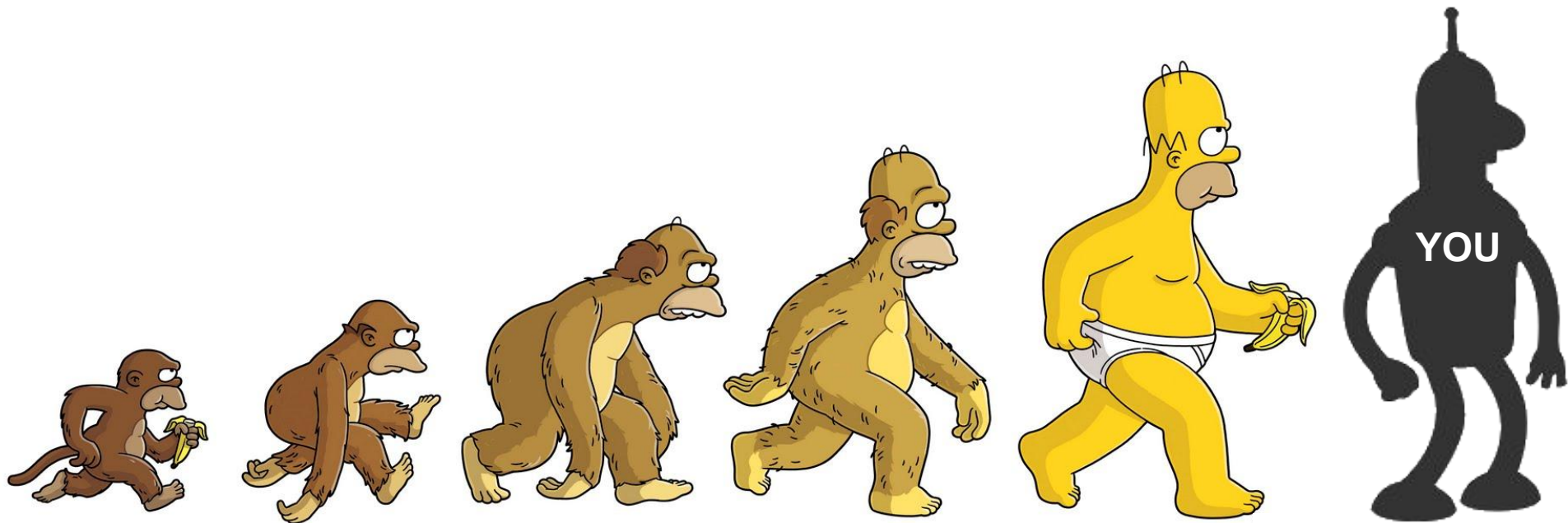
¡Por qué?

Por que no se lo dices?

Quizás lo haga...



EVOLUTION



ONC-RPC
1988 1989

CORBA
1991

RMI
EJB SOAP REST
1996 1997-98 1999 2000

JAX-RS
2008

???
2015

HTTP

WWW

0.9

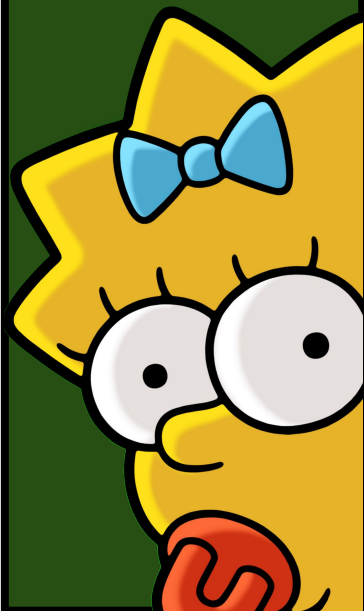
1.0

1.1

2.0

ONE

**RESOURCE
URL**



TWO

**HTTP
METHODS**



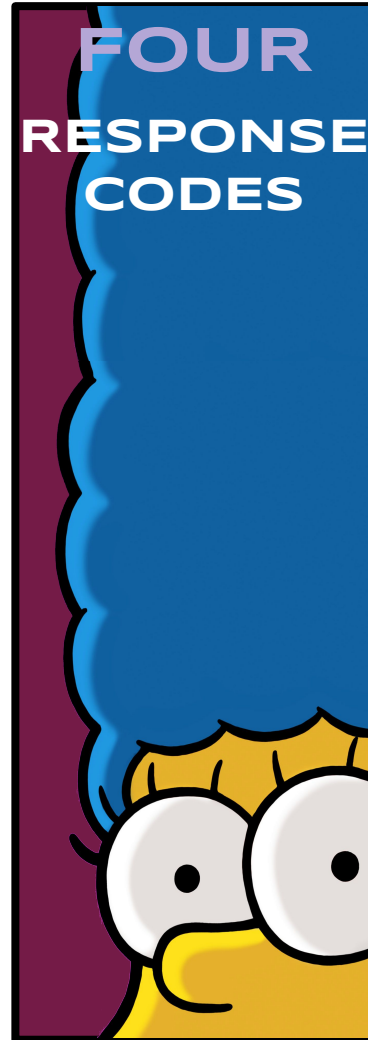
THREE

**MULTIPLE
REPRESENTATIONS**



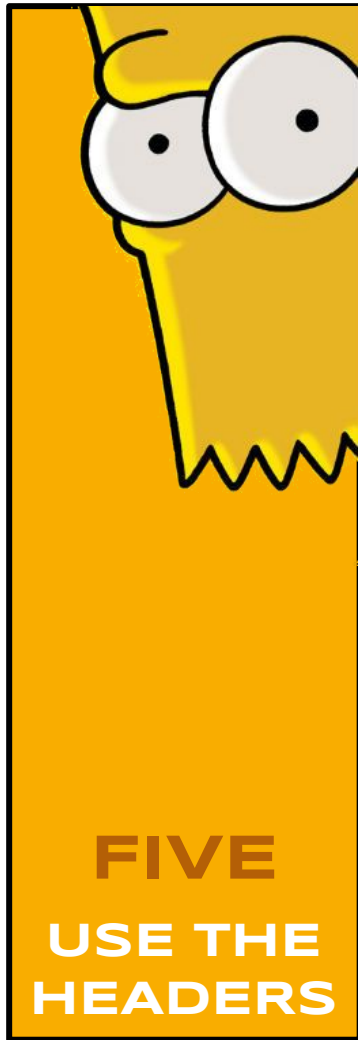
FOUR

**RESPONSE
CODES**



FIVE

**USE THE
HEADERS**





ONE

RESOURCE URL

ASIGNA URLS A TUS RECURSOS

[http://www.simpsons.
com/characters/maggie/father](http://www.simpsons.com/characters/maggie/father)

[/mother](#)

GOOD URL

Fácil de leer/escribir, decir y recordar

Describen el recurso

Utilizan nombres en lugar de verbos

Todos los niveles/paths de la URL son recursos

Asigna un **<id>** (mejor universal) a tus recursos

Filtra recursos mediante **?query-params**

ID Universal: estandarizado/normalizado



<http://www.simpsons.com/characters/>



/homer
/marge
/bart
/lisa
/burns
/moe
/nelson
/skinner
/milhouse

ALIAS vs. IDENTIFIERS

ALIAS:VALUE

/characters/**marion-anthony**
TABLE IDENTIFIER (PK)

/characters/**nickname:fat-**
TABLE ALIAS (UI)
tony

PK: Primary Key

UI: Unique Index



URL PATTERNS

MANY_{TO}ONE · ONE_{TO}MANY

M_{TO}O O_{TO}M M_{TO}O O_{TO}M M_{TO}O

/characters/**maggie**/brothers/**bart**/cousins/**lin**

TABLE PK RELATION PK RELATION PK



URL PATTERNS

ONEtoONE

M_{ToO} O_{ToO} O_{ToO} O_{ToO}
/characters/**maggie**/father/mother/couple
TABLE PK RELATION RELATION RELATION
homer mona

abraham



URL FILTERS

MANY?FILTER

M?F
/characters?surname=Flanders
TABLE WHERE



URL FILTERS

MANY?FILTER

^{M_{TO}O}
/characters/^{O_{TO}M}maggie/^{M?F}brothers?
_{TABLE}sex=female _{PK} _{RELATION} _{WHERE}



URL FILTERS

Operator	Name suffix	Value prefix
equals	name=Lisa	
distinct	surname.not=Flanders	surname=!Flanders
in	age=9&age=10	age=9,10
greather	age.from=10	age=>10
less	age.to=15	age=<15
like	name.contains=a name.starts=a name.ends=a	name=*a* name=a* name=*a
order	order=name&order=-surname	order=name,-surname
paginate	from=20&size=10(&count=true)	



<http://simpsons/characters/homer>

id: **homer**

name: Homer Jay Simpson

age: 38

parent: Abraham Simpson

mother: Mona Simpson

married: Marge Simpson

children: Bart, Lisa, Maggie

brothers: Herb, Abbie



<http://simpsons/characters/bart>



id: bart

name: Bartholomew Simpson

age: 10

parent: Homer Simpson

mother: Marge Simpson

brothers: Lisa, Maggie

cousins: Ling

<http://simpsons/characters/abraham>

ANY_{TO}ONE RELATIONS



/mona/couple



/homer/mother/couple



/herb/father



/marge/father-inlaw



/bart/grandfather



/lisa/father/father



/maggie/mother/couple/father



<http://simpsons/characters/homer/children>

ONEtoMANY RELATIONS



/marge/children



/selma/nephews



/patty/sisters/marge/children



/abraham/grandchildren



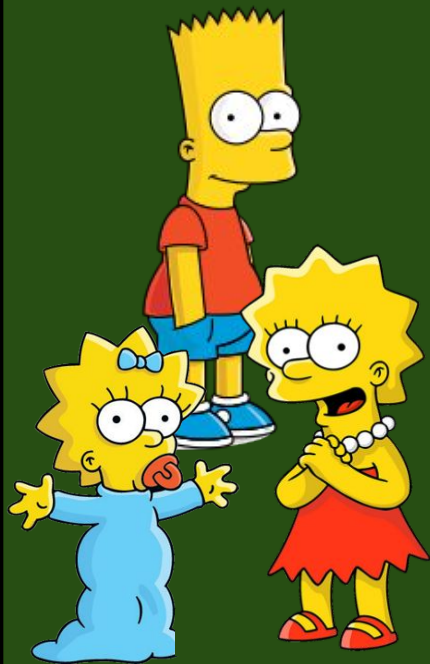
/jackeline/children-



inlaw/homer/children

/ling/cousins

/characters?parent=homer



TWO

HTTP METHODS

INTERACTÚA CON MÉTODOS HTTP

GET /places/moe's-bar/duff



HTTP METHODS

Method	Description	Type
GET	Obtener recurso o colección	safe & idempotent
POST	Crear sin ID (Location) Actualizar uno/más recursos	non-idempotent
PUT	Crear con ID Actualizar completamente	idempotent
DELETE	Eliminar recurso o colección	idempotent
PATCH (*)	Actualizar parcialmente	idempotent

PATCH (*) is proposed, but at moment is not included into HTTP spec



POST METHOD

Usa **POST** cuando no encaje ningún otro método:

GET: obtener/recuperar recursos

PUT: actualizar/reemplazar totalmente un recurso

DELETE: eliminar/borrar un recurso

PATCH: actualizar parcialmente un recurso (*)

Casi todas actualizaciones serán parciales, salvo en ficheros/docs:

PATCH queda desaconsejado hasta que se incluya en HTTP

PUT queda “relegado” a crear recursos con URL/ID conocido

El 99% de las acciones que actualicen tus recursos son POST

POST can't retrieve (GET) or remove (DELETE) resources



SQL SIMILARITIES

HTTP	SQL
GET	SELECT
POST	INSERT / UPDATE
PUT (*)	INSERT / UPDATE
DELETE	DELETE (no DROP)
<ID>	PRIMARY KEY
<ALIAS>:<VALUE>	UNIQUE INDEX
URL/LINK	TABLES/RELATIONS

PUT can INSERT with known ID or UPDATE completely (with all columns)



GET /donuts/1



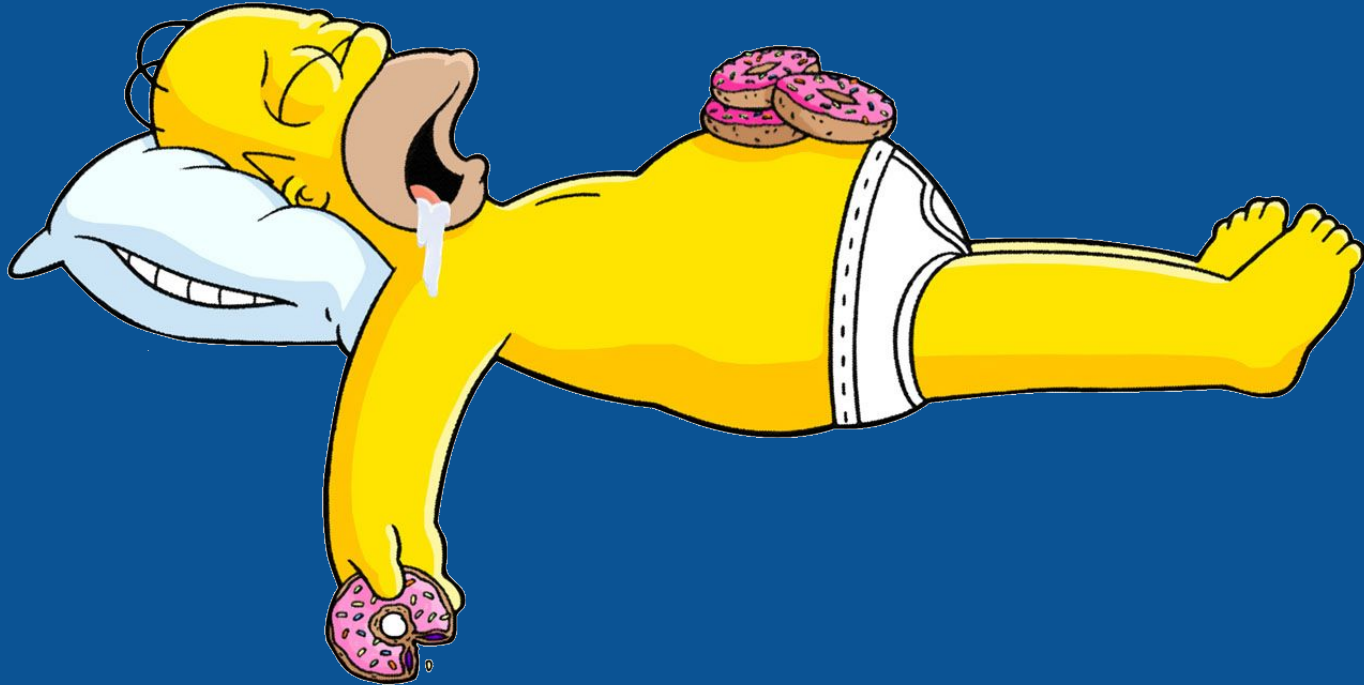
DELETE /donuts/1



GET /donuts



DELETE /donuts



POST /donuts





PUT

/super-donuts/greatwhite



DELETE /super-donuts/greatwhite



THREE

MULTIPLE REPRESENTATIONS

PERMITE MÚLTIPLES
REPRESENTACIONES

XHTML · XML · JSON · CSV · ASCII
PROPERTIES · SVG · PNG · MATHML



MIME TYPES

	Mime-Type
SVG	image/svg+xml
XML	application/xml
JSON	application/json
PROPERTIES	text/plain
CSV	text/csv
TXT	text/plain
XHTML	application/xhtml+xml
MATHML	application/mathml+xml



SVG · SCALABLE VECTOR GRAPHICS

No olvides [enlazar](#)
unos recursos con
otros



XML · EXTENSIBLE MARKUP LANGUAGE



```
<?xml version="1.0" encoding="UTF-8"?>
<character xmlns:xlink="http://www.w3.org/1999/xlink"
  id="lisa"
  desc="Lisa Simpson"
  xlink:href="http://simpsons/characters/lisa">
  <name>Lisa</name>
  <surname>Simpson</surname>
  <nickname>nerd</nickname>
  <parent id="homer" desc="Homer Simpson" xlink:href="/characters/homer"/>
  <mother id="marge" desc="Marge Simpson" xlink:href="/characters/marge"/>
  <brothers xlink:href="/characters/lisa/brothers"/>
  <cousins xlink:href="/characters/lisa/cousins"/>
</character>
```

JSON · JAVASCRIPT OBJECT NOTATION

```
{  "id": "lisa", "desc": "Lisa Simpson", "href": "http://simpsons/characters/lisa",  
  "name": "Lisa",  
  "surname": "Simpson",  
  "nickname": "nerd",  
  "parent": { "id": "homer", "desc": "Homer Simpson", "href": "/characters/homer" },  
  "mother": { "id": "marge", "desc": "Marge Simpson", "href": "/characters/marge" },  
  "brothers": { "/characters/lisa/brothers" },  
  "cousins": { "/characters/lisa/cousins" }  
}
```



PROPERTIES

id=lisa

desc=Lisa Simpson

href=http://simpsons/characters/lisa

name=Lisa

surname=Simpson

nickname=nerd

parent.id=homer

parent.desc=Homer Simpson

parent.href=/characters/homer

mother.id=marge

mother.desc=Marge Simpson

mother.href=/characters/marge

brothers=/characters/lisa/brothers

cousins=/characters/lisa/cousins



CSV · COMMA-SEPARATED VALUES

id,name,surname,parent.id,parent.desc,mother.desc,...

"lisa","Lisa","Simpson","homer","Homer Simpson","Marge Simpson",...

"bart","Bart","Simpson","homer","Homer Simpson","Marge Simpson",...

"maggie","Maggie","Simpson","homer","Homer Simpson","Marge Simpson",...



TXT • ASCII-TEXT

id [31] name [31] surname [31] parent.id [31] parent.desc [31] mother.desc [30]
lisa [31] Lisa [31] Simpson [31] homer [31] Homer Simpson [31] Marge Simpson [30]
bart [31] Bart [31] Simpson [31] homer [31] Homer Simpson [31] Marge Simpson [30]

ASCII DELIMITED TEXT

#	Description
28	FS: File Separator
29	GS: Group Separator
30	RS: Record (row) Separator
31	US: Unit (field) Separator



HTML · HYPERTEXT MARKUP LANGUAGE

GET /characters/**lisa**

```
<form name="character" method="GET" action="/characters/lisa;edit">
```

Identifier: **lisa**

Name: Lisa

Surname: Simpson

Nickname: Nerd

Parent: Homer Simpson

Mother: Marge Simpson

Relations: Brothers,
Cousins

```
<input type="submit" value="Edit">
```

```
</form>
```



HTML

GET /characters/**lisa**;edit

```
<form name="character" method="POST" action="/characters/lisa">  
  Identifier: <input type="text" name="id" value="lisa"/>  
  Name: <input type="text" name="name" value="Lisa"/>  
  Surname: <input type="text" name="surname" value="Simpson"/>  
  Nickname: <input type="text" name="nickname" value="nerd"/>  
  Parent: <input type="text" name="parent.id" value="homer"/>  
  Mother: <input type="text" name="mother.id" value="marge"/>  
  
  <input type="submit" value="Save">  
</form>
```

HTML Forms not allows PUT (neither DELETE)



HTML

GET /characters/new

```
<form name="character" method="POST" action="/characters">  
  Identifier: <input type="text" name="id" value=""/>  
  Name: <input type="text" name="name" value=""/>  
  Surname: <input type="text" name="surname" value=""/>  
  Nickname: <input type="text" name="nick" value=""/>  
  Parent: <input type="text" name="parent.id" value=""/>  
  Mother: <input type="text" name="mother.id" value=""/>  
  
  <input type="submit" value="Create">  
</form>
```



MATHML · MATHEMATICAL MARKUP LANGUAGE

```
<math xmlns="http://www.w3.org/1998/Math/MathML">  
  <mi>E</mi>  
  <mo>=</mo>  
  <mi>m</mi>  
  <msup>  
    <mi>c</mi>  
    <mn>2</mn>  
  </msup>  
</math>
```



MULTIPART

HEADERS	BODY
Content-Type: multipart/mixed boundary=----- 287032381131322 Content-Length: 5540	next rows
-----287032381131322 Content-Disposition: name="homer"; filename="homer.gif" Content-Type: image/gif	
-----287032381131322 Content-Disposition: name="bart"; filename="bart.svg" Content-Type: image/svg+xml	
-----287032381131322 Content-Disposition: name="maggie"; filename="maggie.ico" Content-Type: image/x-icon -----287032381131322--	



A cartoon character with bright blue hair and large, white, round eyes with black pupils. The character is yellow and is looking towards the right side of the frame.

FOUR

RESPONSE CODES

RESPONDE APROPIADAMENTE

Siempre hay que ser educado

RESPONSE CODES

Code	Type
1XX	Informational
2XX	Success
3XX	Redirection
4XX	Client Error
5XX	Server Error



SUCCESS

#	Response	Description	GET	POST	PUT	DEL
200	Ok	Successful (*)	X	X	X	X
201	Created	Successful creation (*)		X	X	
202	Accepted	Request received		X	X	X
204	No Content	Successful without content	X		X	X

(*) Return body contains created Resource



REDIRECTION

#	Response	Description	GET
301/308	Moved Permanently	Resource has moved (*)	X
302/307	Temporary Redirect	Moved Temporary (*)	X
303	See Other	Redirection (*)	X
304	Not Modified	Resource not modified	X

(*) Return Location Header to other Resource



ERROR

#	Response	Description	GET	POST	PUT	DEL
400	Bad Request	Invalid request	X	X	X	X
401	Unauthorized	Authentication needed	X	X	X	X
403	Forbidden	Authentication failure	X	X	X	X
404	Not Found	Resource not found	X	X	X	X
405	Not Allowed	Method not allowed (*)	X	X	X	X
406	Not Acceptable	Representation not available	X			
408	Timeout	Request timeout	X	X		
409	Conflict	State not permit request		X	X	X
415	Unsupported Type	Unsupported representation		X	X	

(*) Method not supported for concrete resource

SERVER ERROR

#	Response	Description	GET	POST	PUT	DEL
500	Server Error	Internal server error	X	X	X	X
501	Not Implemented	Operation not supported (*)		X	X	X
503	Service Unavailable	Temporarily unavailable	X	X	X	X

(*) Method not supported for any resource





Marge
GET /characters/marge

Si?
200 Ok

Te quieres casar conmigo?
POST /characters/marge?marry=artie

No, estoy casada con Homer
409 Conflict

State of the resource doesn't permit request



Esta "Al"?

GET /places/moes-bar?name=Al

Quien?

404 Not found

"Al" de apellido "Reves"

GET /places/moes-bar?name=Al&surname=Revés

Esta "Al Reves"?

Ja,Ja,Ja,Ja

☠ @ ⚡ ☢ ! *

400 Bad Request



Dios creó a Adán y Eva

PUT /characters/adan

PUT /characters/eva

201 Created

Después nacieron
Caín y Abel

PUT /characters/cain

PUT /characters/abel

201 Created

Caín mató a Abel

DELETE /characters/abel

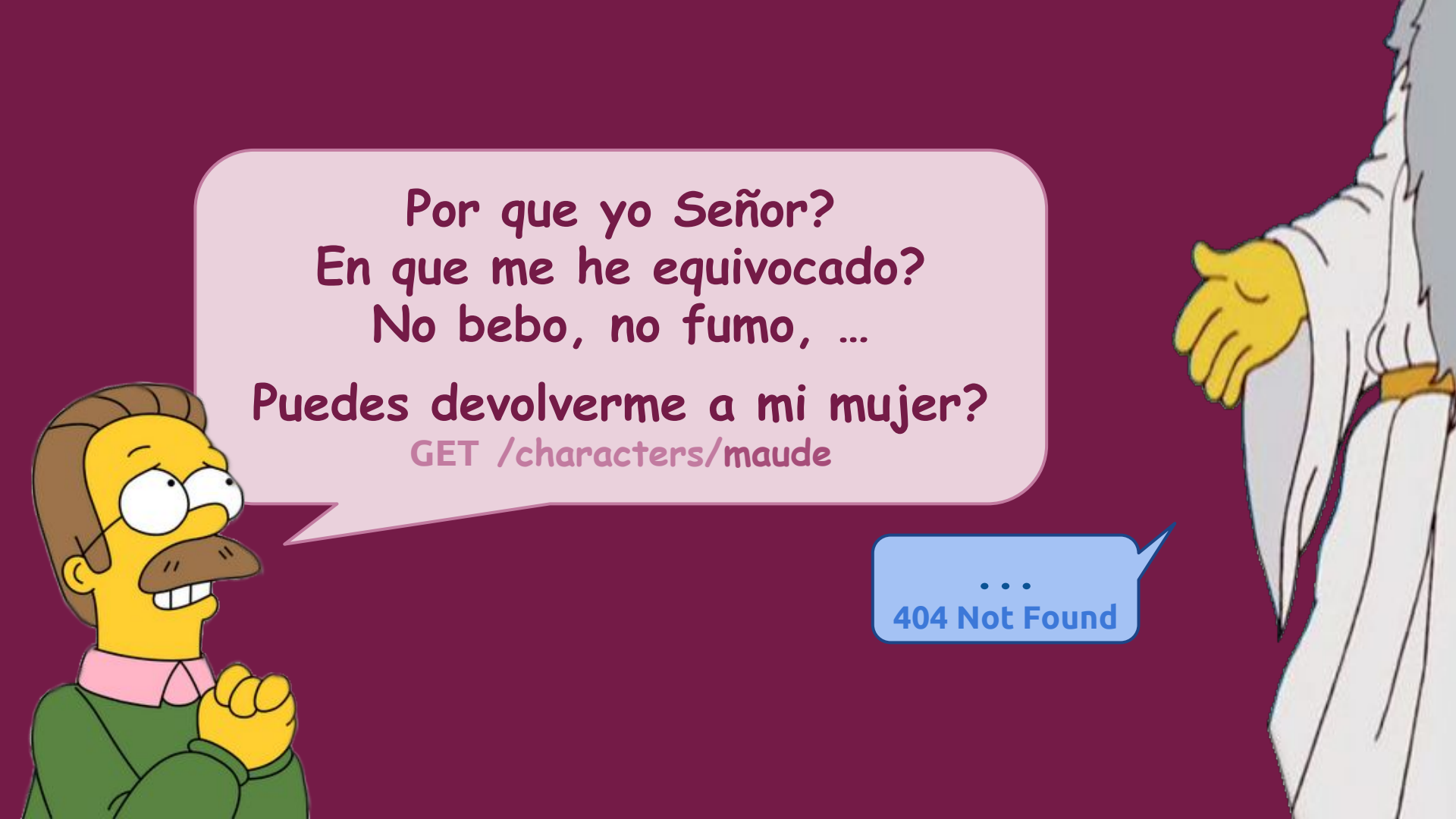
200 Ok

Con estos hechos,
Por que los ateos
niegan a Dios?

DELETE /God

405 Method
Not Allowed





Por que yo Señor?
En que me he equivocado?
No bebo, no fumo, ...
Puedes devolverme a mi mujer?
GET /characters/maude

...
404 Not Found



Duffman, necesito un pedido
de 10 barriles de cerveza

`POST /requests/duff?quantity=10`

Los tendrás esta
misma tarde

202 Accepted
(but not processed)

Pending Request: could be rejected while it's
being processed...





Tienes más Donuts?

GET /apu-store/donut?quantity=10

En la sección de bollería

302/307 Temporary redirect

Y armas de fuego?

GET /apu-store/guns?quantity=1

**No, la policía las
confiscó...**

301/308 Moved permanently



FIVE

USE THE HEADERS

HTTP HEADERS

HTML CLIENTS



REQUEST HEADERS

Header	Description
Accept	Content-Types acceptable for the response
Accept-Charset	Character sets that are acceptable
Accept-Encoding	List of acceptable encodings: gzip, deflate
Accept-Language	List of human languages for response
Authorization	Authentication credentials
Cookie	HTTP cookie previously sent by the server
Content-Type	Mime type of the request body (POST, PUT)



RESPONSE HEADERS

Header	Description
Content-Encoding	Type of encoding used: gzip, deflate
Content-Language	The human language of content
Content-Type	The MIME type of this content
Location	In redirection, or new resource created
Set-Cookie	HTTP cookie
WWW-Authenticate	Authentication scheme to access
Cache-Control	Control cache mechanism



CACHE-CONTROL HEADER

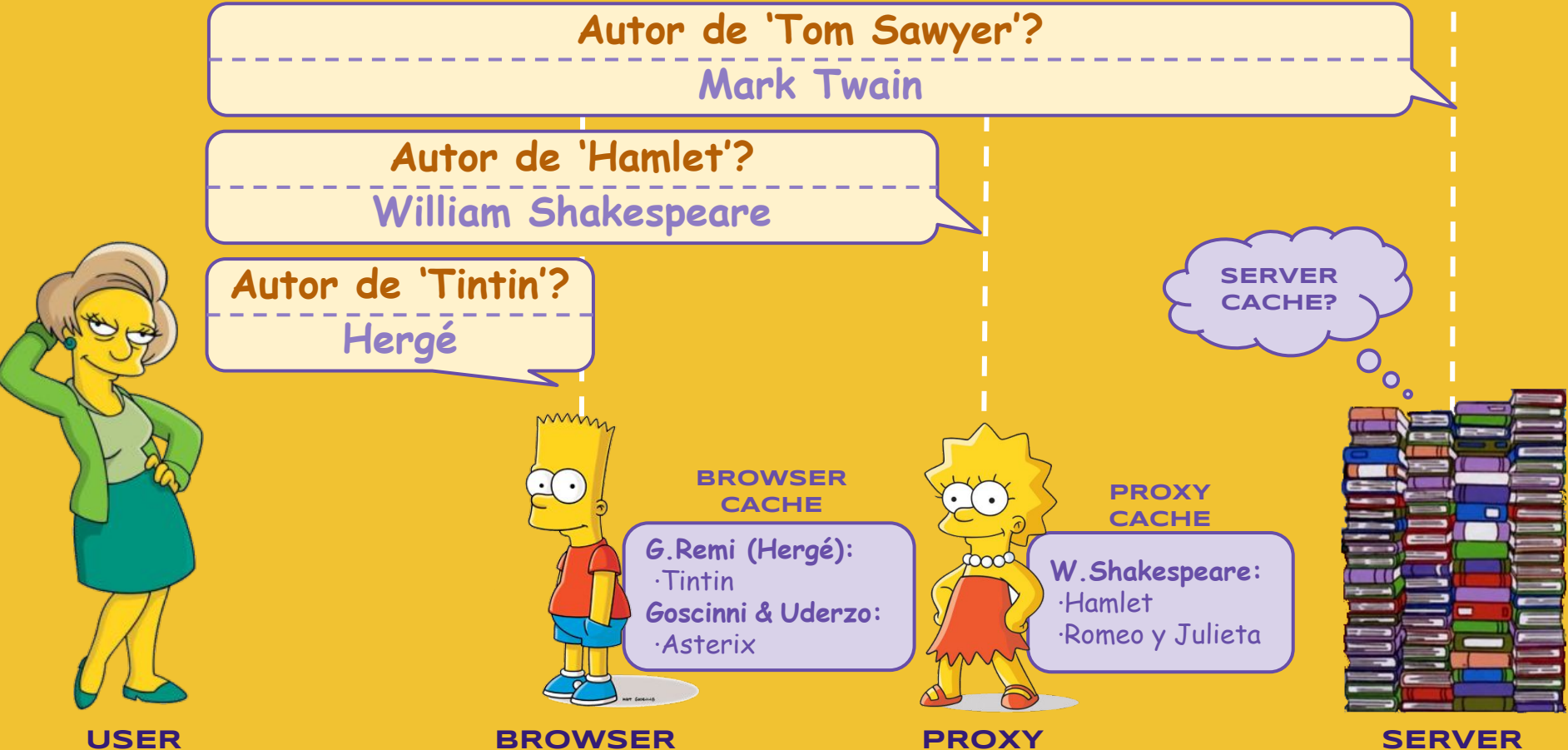
Cache-Control	Action
private	Enable (End)Client cache
public	Enable Proxy/Client cache
no-cache, no-store	Disable cache
public, max-age=X (*)	Enable cache (X <= 31536000sg/1yr)

(*) Enable cache until <date expiration> or <version update> with headers:

- **Expires:** Mon 4 May 2015 21:31:12 GMT (until date expiration)
- **ETag:** 737060cd8c284d8af7ad3082f209582d (until version update)



CACHE MECHANISMS



LANGUAGE SUPPORT



Soy Chino

Accept-Language: zh

你好吗？

Content-Language: zh



Soy Hindú

Accept-Language: hi

आप कैसे हैं?

Content-Language: hi



Soy Sueco

Accept-Language: sv

Hur mår du?

Content-Language: sv



Soy Escocés

Accept-Language: en

How are you?

Content-Language: en



REDIRECT TO LOCATION



Tienes más Donuts?

GET /apu-store/donut?quantity=10

En la sección de bollería

302/307 Temporary redirect

Location: /apu-store/buns-section

Y armas de fuego?

GET /apu-store/guns?quantity=1

No, la policía las confiscó...

Pregunta en la armería

301/308 Moved permanently

Location: /weapons-store/guns



CREATION WITHOUT ID

POST /funzos?color=brown
201 Created
Location: /funzos/1024



POST /funzos?color=blue
201 Created
Location: /funzos/1022



POST /funzos?
-----color=purple-----
201 Created
Location: /funzos/1020



POST /funzos?color=green
201 Created
Location: /funzos/1023



POST /funzos?color=red
201 Created
Location: /funzos/1021



CREATE WITH IDENTIFIER

Soy
radioactiveman

Soy
pieman

Soy
burnsman

Superhéroes o Superzeroes?

Burnsman, otro millonario
imitador de Bruce Wayne
PUT /zeroes/**burnsman**

Pieman, traje casero
y lanzador de tartas
PUT /zeroes/**pieman**

Radioactiveman, un auténtico
superhéroe radioactivo...
PUT /heroes/**radioactiveman**



BASIC SECURITY



Toc, toc...
<No header Authorization>

Quien es?

401 Unauthorized
WWW-Authenticate: Basic

policia:abra

Authorization: Basic cG9saWNpYTphYnJh

Lo siento, sólo socios

403 Forbidden



COMPRESSION



Si fuera pequeñito las
rosquillas durarían más

GET /characters/homer

Accept-Encoding: gzip

Yuuuuuuuuuuuu!! Tengo
rosquillas para 2 semanas

Content-Encoding: gzip



CONDITIONAL REQUESTS



Header	Description
ETag	<u>Version</u> of a resource, often a message digest
If-Match (PUT)	Optimistic Locking: Only update a resource if 'etag' has not been modified else return 412 (Precondition Failed)
If-None-Match (GET)	Allows a 304 (Not Modified) if 'etag' is unchanged
Last-Modified	The last modified <u>date</u> for the requested object
If-Unmodified-Since (PUT)	Optimistic Locking: Only update a resource if 'date' has not been modified else return 412 (Precondition Failed)
If-Modified-Since (GET)	Allows a 304 (Not Modified) if 'date' is unchanged

ETag: 737060cd8c284d8af7ad3082f209582d

Last-Modified: Tue, 12 May 2015 12:45:26 GMT

CONDITIONAL GET

Tienes un dolar?
GET /milhouse/money

Si
200 Ok

ETag: 258df1f625

Last-Modified: Tue, 12 May 2015 12:45:26 GMT

Que si tienes un dolar!

GET /milhouse/money

If-None-Match: 258df1f625

If-Modified-Since: Tue, 12 May 2015 12:45:26 GMT

Ya te he dicho que Si
304 Not Modified (No body)

Grrr.. Dame todo tu dinero!

DELETE /milhouse/money

If-None-Match: 258df1f625

If-Modified-Since: Tue, 12 May 2015 12:45:26 GMT



CONDITIONAL PUT

Tienes novio?

GET /lisa

No tengo

200 Ok

ETag: 12df1f5

Last-Modified: Tue, 12
May 2015 12:45:26 GMT

Quieres ser mi novia?

PUT /lisa?boyfriend=milhouse

If-Match: 12df1f5

If-Unmodified-Since: Tue,
12 May 2015 12:45:26 GMT

Vale

200 Ok

Quieres ser mi novia?

PUT /lisa?boyfriend=nelson

If-Match: 12df1f5

If-Unmodified-Since: Tue,
12 May 2015 12:45:26 GMT

Ya tengo novio

412 Precondition Failed



ONE

ASIGNA URLS A TUS RECURSOS

`http://simpsons/characters/bart/donations`



TWO

INTERACCIONA CON MÉTODOS HTTP

`POST http://simpsons/characters/bart/donations`



THREE

PERMITE MÚLTIPLES REPRESENTACIONES

`xml: <money currency="USD">0</money>`
`json: {"money": {"amount": 1,"currency": "EUR"}}`
`form: }`
`money=5¤cy=GBP`



FOUR

RESPONDE APROPIADAMENTE

Tacaño: 400 Bad Request (money is 0)
Ladrón: 405 Method Not Allowed (DELETE)
Gracias: 200 OK



FIVE

USA LAS CABECERAS

Seguridad, Multidioma, Redirección
Cache, Conditional Requests,

Formatos



*Un pesimista ve dificultad
en cada oportunidad.*

*Un optimista ve
oportunidad
en cada dificultad.*

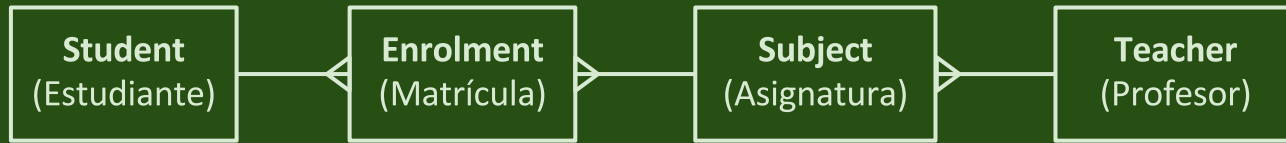
Winston Churchill...





/countries/**ES**/regions/**madrid**/cities/**getafe**?region.country.language=es
/cities/**getafe**/region/country?regions...cities...areas...





Wich are the subjects and teacher of **Lisa** enrollments on **2015-2016**?

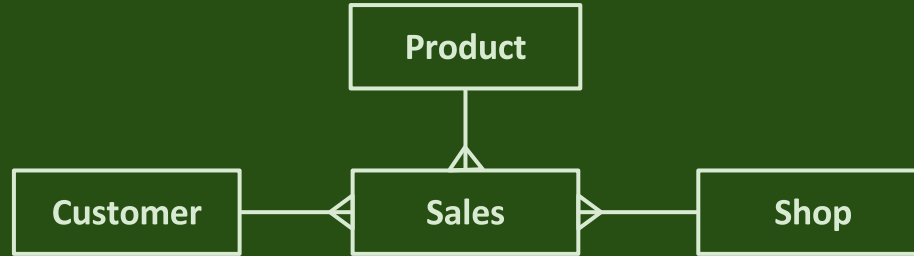
```
/students/lisa/enrollments/2015-2016/subjects;teacher  
/subjects;teacher?enrollments.id=2015-2016&enrollments.student.id=lisa
```

Wich are the students of teacher **Elsa** for subject **english-a1** on **2015-2016**?

```
/teachers/elsa/subjects/english-a1/enrollments/2015-2016/students  
/students?enrollments.id=2015-2016&enrollments.subject.id=english-a1&enrollments.subject.teacher.id=elsa
```

Wich are the enrollments on **2015-2016** sorter by subject.name and student.surname (desc)?

```
/enrollments?enrollments.course=2015-2016&order=subject.name&order=-student.surname
```



Wich are the subjects and teacher of **Lisa** enrollments on **2015-2016**?

/customers/**john**/sales/**20150229000001**/products
/subjects;teacher?enrollments.id=**2015-2016**&enrollments.student.id=**lisa**

Wich are the students of teacher **Elsa** for subject **english-a1** on **2015-2016**?

/teachers/**elsa**/subjects/**english-a1**/enrollments/**2015-2016**/students
/students?enrollments.id=**2015-2016**&enrollments.subject.id=**english-a1**&enrollments.subject.teacher.id=**elsa**

Wich are the enrollments on **2015-2016** sorter by subject.name and student.surname (desc)?

/enrollments?enrollments.course=**2015-2016**&order=subject.name&order=-student.surname



HTTP

+ 20 YEARS



Spec	Version	Year
CERN	0.*	1991
RFC 1945	1.0	1996
RFC 2616	1.1	1999
RFC 2774	1.2	2000
DRAFT	2.0	¿2015?

*La primera señal de que estás
innovando es que la gente no
te entiende...*

Ferran Adrià



ALIAS vs. IDENTIFIERS

ALIAS:VALUE

/robots/**bender**

TABLE

PK

/robots/**unit:22**

TABLE

UI

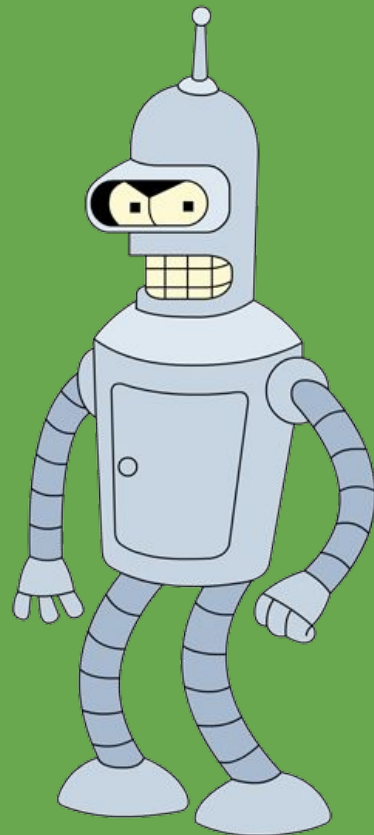
/robots/**serial:2716057**

TABLE

UI

PK: Primary Key

UI: Unique Index



<http://www.simpsons.com/zeroes/>
(superheroes - superpowers = zeroes)

/burnsman



/everyman



/bartman



/falloutboy





edwina



louis



abraham



mona



jackeline



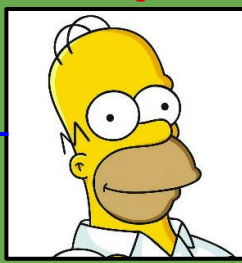
clancy



abbie



herb



homer



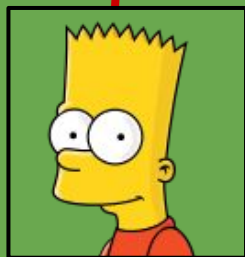
marge



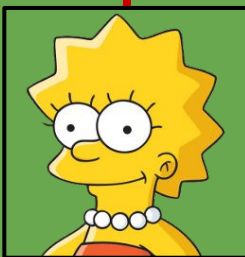
patty



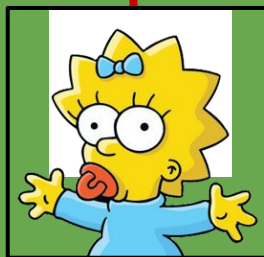
sel ma



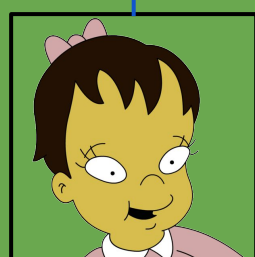
bart



lisa



maggie



ling

<http://simpsons/characters/marge/sisters>



[/homer/sisters-in-law](#)



[/bart/aunts](#)



[/lisa/mother/sisters](#)



[/maggie/father/sisters-in-law](#)



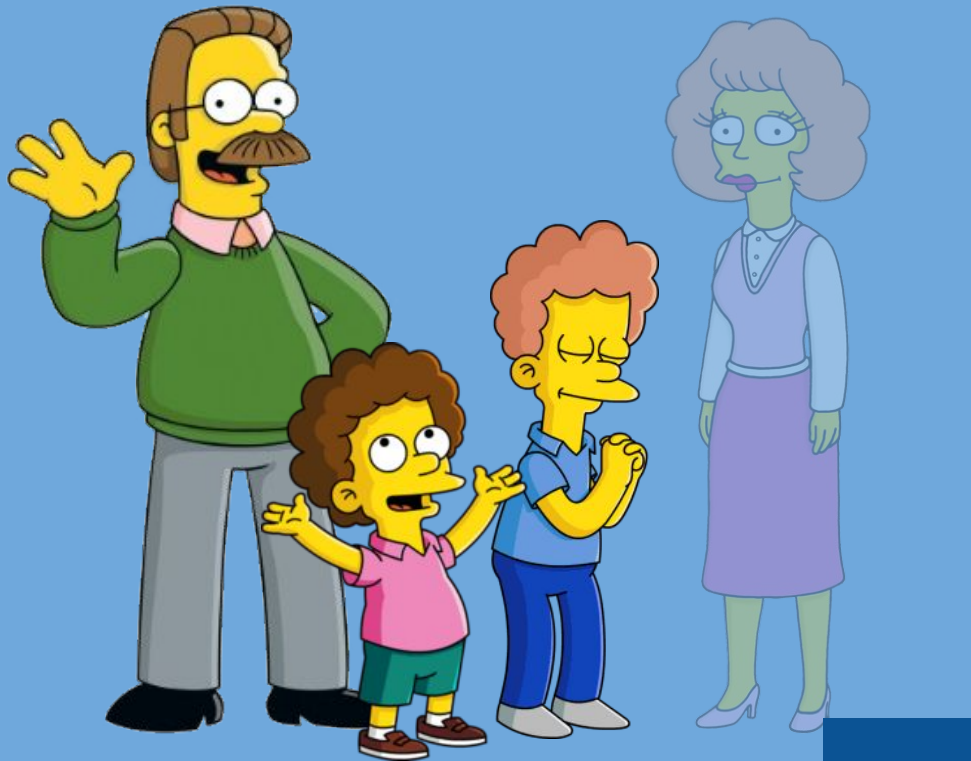
[/clancy/children?surname=Bouvier](#)



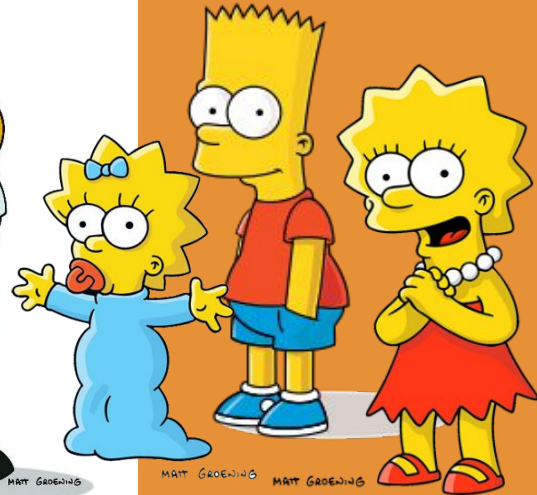
[/jackeline/children?status=single](#)



DELETE /characters/maude



GET /simpsons/characters?surname=Simpson



DELETE /characters/**bart**



Un claro caso
de accidente
doméstico
(mira que caerse
sobre el cuchillo)



DELETER LICENSE

SPRINGFIELD

EXP. 007



NAME: JAMES BOND

CLASS: SECRET AGENT

EXPIRES: 2017.07.07

SIGNATURE: JAMES BOND

COMPRESSION



Vamos a China por
nuestro aniversario?



Cuanto vale el billete para
China? Y cada maleta?

```
GET /air-china/prices  
Accept-Encoding: gzip
```



Vale

✈ AIR CHINA ✈

Pasajeros: \$1200

Maletas: \$15

Content-Encoding: gzip

HTML CLIENTS

form.action: only allows GET, POST (ni PUT, ni DELETE)

links: can't send headers request

- mime-types (.extension)

creation form: GET /resources/new

edit forms: GET /resources/<id>;edit (;view)

Post-Redirect-Get: avoid re-post on reloads

disable buttons on submit: avoid re-submit on double click

url aliases: transform url

- subresources as resources & path to query params

polimorphic urls:

- /orders/periodic-transfers?date=today
- /orders/transfers?type=periodic&date=today



SUMMARY

1. Asigna URLs a tus recursos

`http://simpsons/characters/bart`

2. Interacciona con métodos HTTP

`POST http://simpsons/characters/bart/donations`

3. Permite múltiples Representaciones

```
<money currency='USD'>0</money>
{'money':{'amount':1,'currency':'EUR'}}
money=5&currency=GBP
```

4. Responde apropiadamente

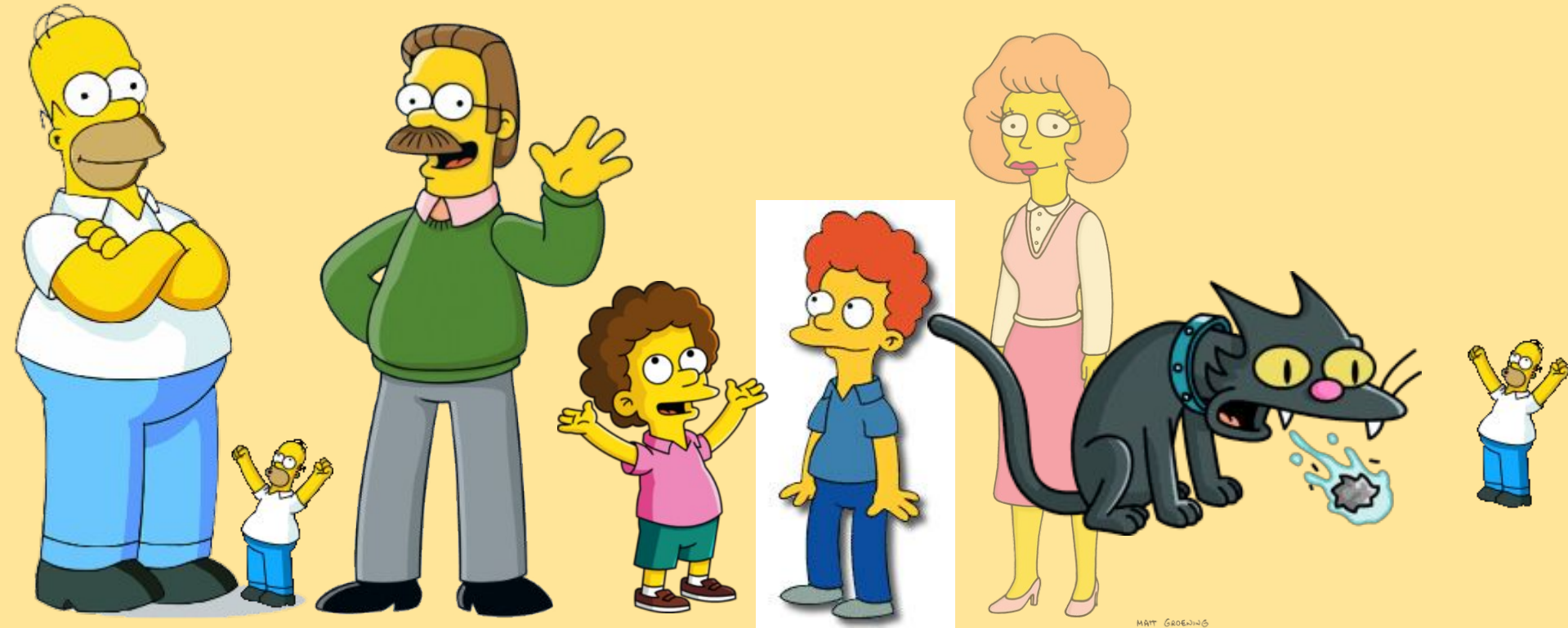
```
400 Bad Request (money is 0)
405 Method Not Allowed (GET)
200 OK
```

5. “Abraza” el protocolo HTTP

Usa cabeceras HTTP: Seguridad, Multidioma, Redirección
Cache, Conditional Requests, Formatos



GET /simpsons?surname=Flanders

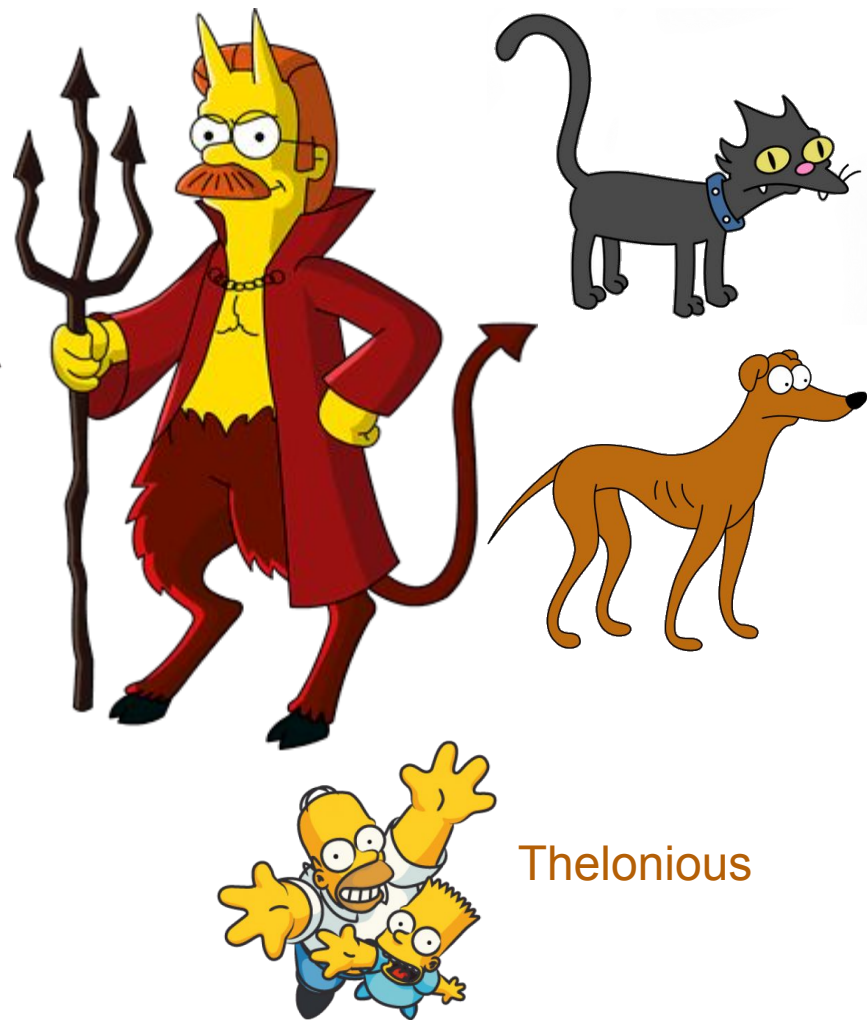


DELETE <http://simpsons/rasca>





Edmund



Thelonious



<http://simpsons>





jeff

GET <http://simpsons/places/moe's>





clobber girl



the-collector



blue-falcon



fallout-boy



bat-man-fruit



radiative-man



everyman



cupcake-kid

pieman
cupcakekid



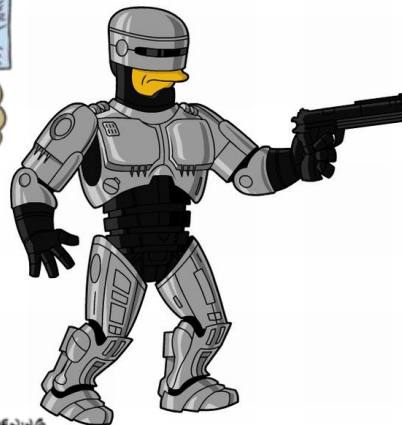
strech-dude

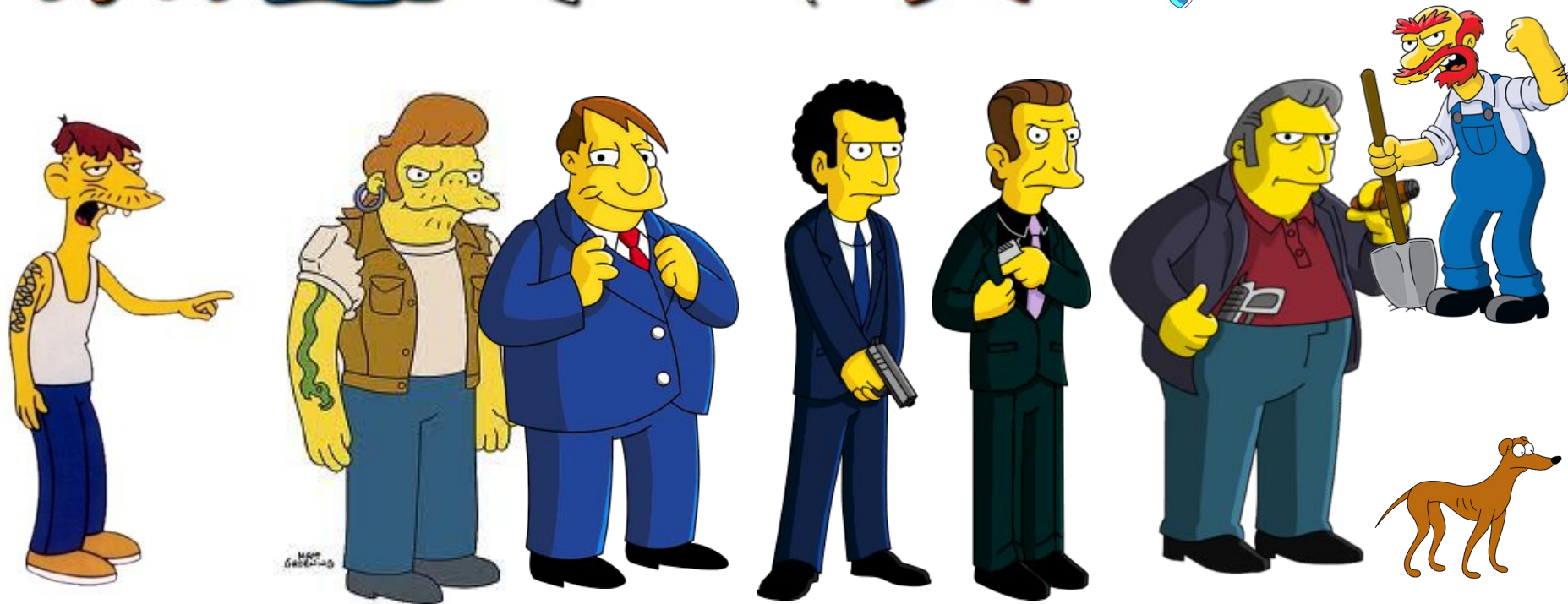


I'm the
Goddamn
Bartman

bartman

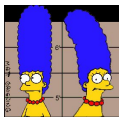
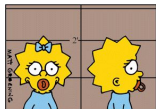
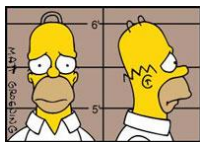
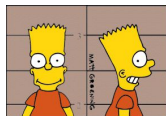


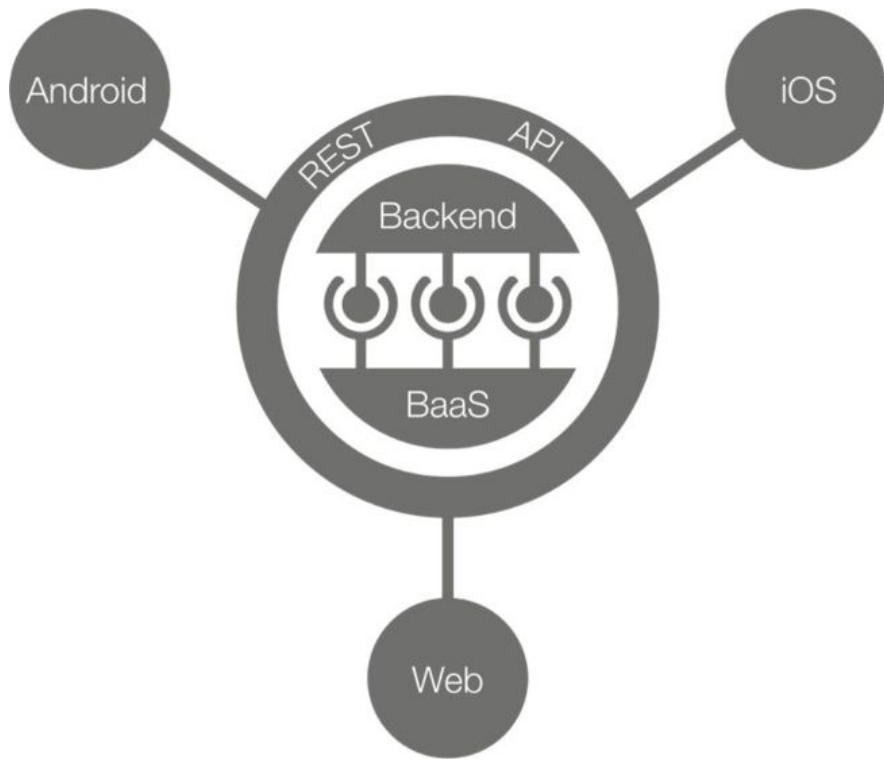




THE SIMPSONS







http://es.wikipedia.org/wiki/Backend_as_a_service